

Two Beliefs: the Proofs

One residual under two beliefs about healthy data

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Setup

window to score $\mathbf{y} \in \mathbb{R}^L$

belief about healthy windows $X \sim \mu$

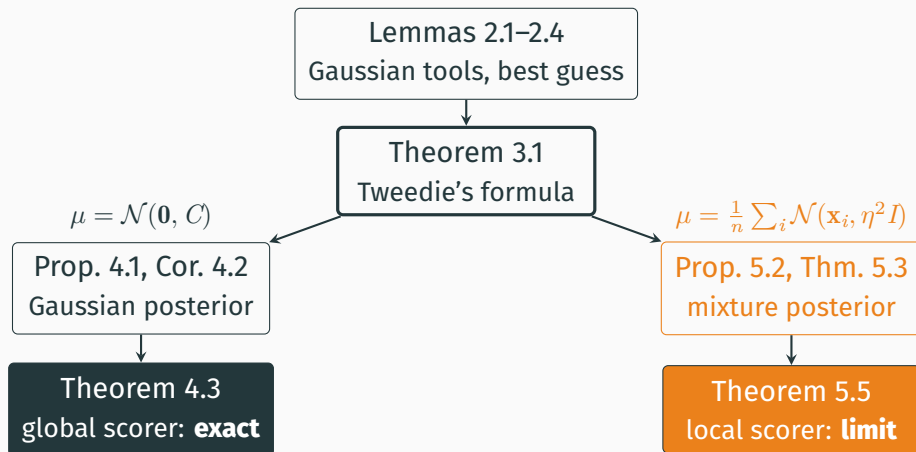
observation $Y = X + \sqrt{\varepsilon} Z$, $Z \sim \mathcal{N}(\mathbf{0}, I)$ independent of X

training windows $\mathbf{x}_1, \dots, \mathbf{x}_n$, $C = \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i^\top = \sum_j \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$

$$a_{\mu, \varepsilon}(\mathbf{y}) = \left\| \mathbf{y} - \mathbb{E}_{\mu}[X \mid Y = \mathbf{y}] \right\|$$

denoising residual: distance from the best healthy guess

The chain of results



$$\phi_s(\mathbf{v}) = (2\pi s)^{-L/2} \exp(-\|\mathbf{v}\|^2/2s) \quad \text{density of } \mathcal{N}(\mathbf{0}, sI)$$

Lemma 2.1 gradient $\nabla_{\mathbf{y}} \phi_s(\mathbf{y} - \mathbf{x}) = \frac{\mathbf{x} - \mathbf{y}}{s} \phi_s(\mathbf{y} - \mathbf{x})$

Lemma 2.2 density of Y $p_Y(\mathbf{y}) = \int \phi_\varepsilon(\mathbf{y} - \mathbf{x}) \mu(d\mathbf{x}) > 0$, smooth, any μ

Lemma 2.3 posterior $\mu(d\mathbf{x} \mid \mathbf{y}) = \frac{\phi_\varepsilon(\mathbf{y} - \mathbf{x}) \mu(d\mathbf{x})}{p_Y(\mathbf{y})}$

$$f^*(\mathbf{y}) = \mathbb{E}[X \mid Y = \mathbf{y}] \text{ minimises } \mathbb{E}\|f(Y) - X\|^2$$

fix $\mathbf{y}$, $\mathbf{m} = \mathbb{E}[X \mid \mathbf{y}]$, any $\mathbf{a}$:

$$\mathbb{E}[\|\mathbf{a} - X\|^2 \mid \mathbf{y}] = \|\mathbf{a} - \mathbf{m}\|^2 + \mathbb{E}[\|\mathbf{m} - X\|^2 \mid \mathbf{y}] + 2(\mathbf{a} - \mathbf{m})^\top \underbrace{\mathbb{E}[\mathbf{m} - X \mid \mathbf{y}]}_{=0}$$

smallest at $\mathbf{a} = \mathbf{m}$, for every $\mathbf{y}$

$$\mathbb{E}[X \mid Y = \mathbf{y}] = \mathbf{y} + \varepsilon \mathbf{s}(\mathbf{y}), \quad \mathbf{s} = \nabla \log p_Y$$

$$1 \quad \nabla p_Y(\mathbf{y}) = \int \frac{\mathbf{x} - \mathbf{y}}{\varepsilon} \phi_\varepsilon(\mathbf{y} - \mathbf{x}) \mu(d\mathbf{x}) \quad \text{Lemmas 2.1, 2.2}$$

$$2 \quad \mathbf{s}(\mathbf{y}) = \frac{\nabla p_Y}{p_Y} = \frac{1}{\varepsilon} \int (\mathbf{x} - \mathbf{y}) \mu(d\mathbf{x} \mid \mathbf{y}) = \frac{1}{\varepsilon} (\mathbb{E}[X \mid \mathbf{y}] - \mathbf{y}) \quad \text{Lemma 2.3}$$

3 multiply by ε , rearrange

$$\text{so } a_{\mu, \varepsilon}(\mathbf{y}) = \varepsilon \|\mathbf{s}(\mathbf{y})\|$$

$\mu = \mathcal{N}(\mathbf{m}, S), \quad S \succ 0 \quad \implies \quad X \mid \mathbf{y} \text{ Gaussian with}$

$$\mathbb{E}[X \mid \mathbf{y}] = \mathbf{m} + S(S + \varepsilon I)^{-1}(\mathbf{y} - \mathbf{m}), \quad \text{Cov}[X \mid \mathbf{y}] = \varepsilon S(S + \varepsilon I)^{-1}$$

1 $\log \text{posterior} = -\frac{1}{2}(\mathbf{x} - \mathbf{m})^\top S^{-1}(\mathbf{x} - \mathbf{m}) - \frac{1}{2\varepsilon}\|\mathbf{y} - \mathbf{x}\|^2 + \text{const}$

$$= -\frac{1}{2}\mathbf{x}^\top P\mathbf{x} + \mathbf{x}^\top \mathbf{b} + \text{const}, \quad P = S^{-1} + \frac{1}{\varepsilon}I, \quad \mathbf{b} = S^{-1}\mathbf{m} + \frac{1}{\varepsilon}\mathbf{y}$$

2 quadratic in $\mathbf{x} \implies$ Gaussian, covariance P^{-1} , mean $P^{-1}\mathbf{b}$

$$3 \quad P = S^{-1}(S + \varepsilon I)/\varepsilon \Rightarrow P^{-1} = \varepsilon(S + \varepsilon I)^{-1}S = \varepsilon S(S + \varepsilon I)^{-1}$$

$$4 \quad P^{-1}\mathbf{b} = \varepsilon(S + \varepsilon I)^{-1}\mathbf{m} + S(S + \varepsilon I)^{-1}\mathbf{y} = \mathbf{m} + S(S + \varepsilon I)^{-1}(\mathbf{y} - \mathbf{m})$$

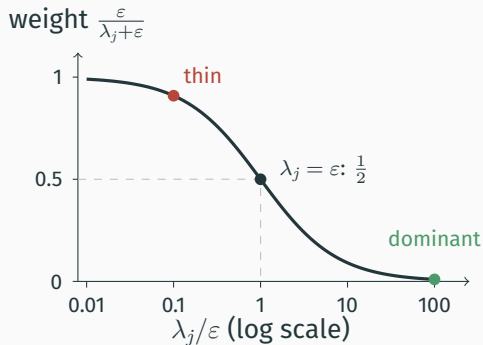
using $\varepsilon(S + \varepsilon I)^{-1} = I - S(S + \varepsilon I)^{-1}$

Corollary 4.2 residual

$$\mathbf{y} - \mathbb{E}[X \mid \mathbf{y}] = \varepsilon (S + \varepsilon I)^{-1}(\mathbf{y} - \mathbf{m})$$

$$a_{\mu,\varepsilon}(\mathbf{y})^2 = \sum_{j=1}^L \left(\frac{\varepsilon}{\lambda_j + \varepsilon} \right)^2 a_j^2,$$

$$a_j = \mathbf{u}_j^\top (\mathbf{y} - \mathbf{m})$$



worked example, $\varepsilon = 1$

λ_j	weight	a_j	contrib.
100	0.0099	10	0.099
100	0.0099	16	0.158
0.1	0.91	1	0.91

Theorem 4.3: the global scorer is this residual

proved

checked

$$P^{(\gamma)} = \sum_{j=1}^L \frac{\gamma^2}{\lambda_j + \gamma^2} \mathbf{u}_j \mathbf{u}_j^\top, \quad \gamma^2 = \lambda_{d+1}, \quad \lambda_j \text{ of the uncentred } C$$

$$\|P^{(\gamma)} \mathbf{y}\| = a_{\mathcal{N}(\mathbf{0}, C), \gamma^2}(\mathbf{y})$$

proof: Cor. 4.2 with $S = C$, $\mathbf{m} = \mathbf{0}$, $\varepsilon = \gamma^2$ gives $\gamma^2(C + \gamma^2 I)^{-1} \mathbf{y}$; Fact 1.5 turns it into $P^{(\gamma)} \mathbf{y}$

largest relative difference, every window

$$7.0 \times 10^{-15} \quad 1.8 \times 10^{-13} \quad 2.3 \times 10^{-15}$$

012_NAB

560_YAHOO

842_OPPORTUNITY

The belief is chosen, not fitted

- training windows treated as clean, with covariance C ; test windows get extra noise γ^2
- self-consistent prior: covariance $C - \gamma^2 I$, clipped at 0 (as in probabilistic PCA)

Zero mean, uncentred C

$C = \Sigma + \bar{\mathbf{x}} \bar{\mathbf{x}}^\top \Rightarrow$ mean direction = one more high-variance direction

864_OPPORTUNITY: VUS-PR **0.07** $\rightarrow$ **0.95** when centred

direction $d + 1$: $\frac{\gamma^2}{\lambda_{d+1} + \gamma^2} = \frac{1}{2}$ exactly

Corollary 4.4 $\lambda_d \gg \varepsilon \gg \lambda_{d+1} \Rightarrow \varepsilon(C + \varepsilon I)^{-1} \approx I - U_d U_d^\top$
the hard subspace residual

but $\varepsilon = \lambda_{d+1}$ breaks $\varepsilon \gg \lambda_{d+1}$

tail spectrum	tail weights	soft vs hard	Spearman
fast decay	≈ 1 (except $d+1$)	close	0.993
flat noise floor	$\approx \frac{1}{2}$ (0.50–0.58)	$\approx$ half	0.994

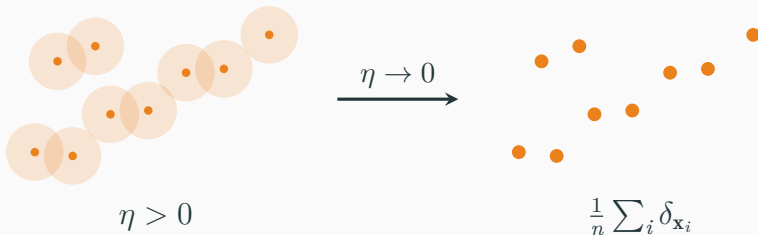
ranking agrees; values do not

$$\frac{\varepsilon}{\lambda_j + \varepsilon} \xrightarrow{\varepsilon \rightarrow \infty} 1 \quad \implies \quad a \rightarrow \|\mathbf{y}\| \quad \text{distance from the belief's mean } 0$$

$$\frac{1}{\varepsilon} \cdot \frac{\varepsilon}{\lambda_j + \varepsilon} \xrightarrow{\varepsilon \rightarrow 0} \frac{1}{\lambda_j} \quad \implies \quad \frac{a}{\varepsilon} \rightarrow \|C^{-1}\mathbf{y}\| \quad (C \text{ invertible})$$

checked: relative difference 10^{-7} at large ε ; Spearman 1.0 with $\|C^{-1}\mathbf{y}\|$ at small ε

$$\mu_\eta = \frac{1}{n} \sum_{i=1}^n \mathcal{N}(\mathbf{x}_i, \eta^2 I)$$



$$\phi_{\eta^2}(\mathbf{x} - \mathbf{x}_i) \phi_{\varepsilon}(\mathbf{y} - \mathbf{x}) = \underbrace{\phi_{\eta^2 + \varepsilon}(\mathbf{y} - \mathbf{x}_i)}_{\text{marginal of } Y} \underbrace{\phi_{s^2}(\mathbf{x} - \boldsymbol{\mu}_i(\mathbf{y}))}_{X|\mathbf{y}}$$

$$\boldsymbol{\mu}_i(\mathbf{y}) = \mathbf{x}_i + \frac{\eta^2}{\eta^2 + \varepsilon}(\mathbf{y} - \mathbf{x}_i), \quad s^2 = \frac{\eta^2 \varepsilon}{\eta^2 + \varepsilon}$$

joint = marginal $\times$ conditional; conditional from Prop. 4.1 with $S = \eta^2 I$, $\mathbf{m} = \mathbf{x}_i$

$$\mathbb{E}[X | \mathbf{y}] = \sum_{i=1}^n r_i(\mathbf{y}) \boldsymbol{\mu}_i(\mathbf{y}), \quad r_i(\mathbf{y}) \propto \exp\left(-\frac{\|\mathbf{y} - \mathbf{x}_i\|^2}{2(\eta^2 + \varepsilon)}\right), \quad \sum_i r_i = 1$$

- 1 posterior $\propto \frac{1}{n} \sum_i \phi_{\eta^2}(\mathbf{x} - \mathbf{x}_i) \phi_{\varepsilon}(\mathbf{y} - \mathbf{x})$ Lemma 2.3
- 2 $= \frac{1}{n} \sum_i \phi_{\eta^2 + \varepsilon}(\mathbf{y} - \mathbf{x}_i) \phi_{s^2}(\mathbf{x} - \boldsymbol{\mu}_i)$ Lemma 5.1
- 3 weights $r_i \propto \phi_{\eta^2 + \varepsilon}(\mathbf{y} - \mathbf{x}_i)$; constants cancel softmax
- 4 mean of a mixture = weighted mean of component means

$$\mathbf{y} - \mathbb{E}[X \mid \mathbf{y}] = \frac{\varepsilon}{\eta^2 + \varepsilon} \left(\mathbf{y} - \sum_i r_i(\mathbf{y}) \mathbf{x}_i \right)$$

$$1 \quad \mathbf{y} - \sum_i r_i \boldsymbol{\mu}_i = \sum_i r_i (\mathbf{y} - \boldsymbol{\mu}_i) \quad (\sum_i r_i = 1)$$

$$2 \quad \mathbf{y} - \boldsymbol{\mu}_i = \left(1 - \frac{\eta^2}{\eta^2 + \varepsilon} \right) (\mathbf{y} - \mathbf{x}_i) = \frac{\varepsilon}{\eta^2 + \varepsilon} (\mathbf{y} - \mathbf{x}_i)$$

$$3 \quad \text{sum over } i, \text{ use } \sum_i r_i = 1 \text{ again}$$

3-million-sample Monte Carlo posterior: agrees to 2–3 decimals (rerun), 3–4 (1 Oct)

$$\eta^2 \rightarrow 0: \quad \mathbf{y} - \mathbb{E}[X \mid \mathbf{y}] \rightarrow \mathbf{y} - \sum_i w_i \mathbf{x}_i, \quad w_i = \text{softmax}_i(-\|\mathbf{y} - \mathbf{x}_i\|^2 / 2\varepsilon)$$

unique nearest window $\mathbf{x}_{(1)}$, $\eta^2/\varepsilon \rightarrow 0$, $\varepsilon \rightarrow 0$:

$$a(\mathbf{y}) = \left\| \mathbf{y} - \sum_i w_i \mathbf{x}_i \right\| \rightarrow \|\mathbf{y} - \mathbf{x}_{(1)}\|$$

- 1 $\frac{w_i}{w_{(1)}} = \exp\left(-\frac{\|\mathbf{y} - \mathbf{x}_i\|^2 - \|\mathbf{y} - \mathbf{x}_{(1)}\|^2}{2\varepsilon}\right), \quad i \neq (1)$
- 2 **bracket** > 0 (uniqueness) $\Rightarrow$ **ratio** $\rightarrow 0$
- 3 $w_{(1)} \rightarrow 1, \quad \sum_i w_i \mathbf{x}_i \rightarrow \mathbf{x}_{(1)}$



illustration: 5 windows, $\|y - x_i\|^2 = 1.0, 1.3, 1.8, 2.5, 3.0$;

nearest in orange

median relative difference to 1-NN

at $\epsilon = 10^{-3} \delta^2$

012_NAB 2.1×10^{-3}

560_YAHOO 2.6×10^{-8}

842_OPPOINT. 7.7×10^{-6}

factor in Theorem 5.3: $\frac{\varepsilon}{\eta^2 + \varepsilon}$

as $\varepsilon \rightarrow 0$	factor	residual norm
$\eta > 0$ fixed	$\rightarrow 0$	$\rightarrow 0$
$\eta^2 = \varepsilon$	$= \frac{1}{2}$	$\rightarrow \frac{1}{2} \ \mathbf{y} - \mathbf{x}_{(1)}\ $
$\eta^2/\varepsilon \rightarrow 0$	$\rightarrow 1$	$\rightarrow \ \mathbf{y} - \mathbf{x}_{(1)}\ $ local scorer

$\eta^2 = 10^{-3}, \varepsilon = 10^{-6}$: 2.3×10^{-4} vs 1-NN 0.233

ties for nearest: the guess tends to their average

$\varepsilon \rightarrow \infty$: weights $\rightarrow \frac{1}{n} \Rightarrow a(\mathbf{y}) \rightarrow \|\mathbf{y} - \bar{\mathbf{x}}\|$ distance from the training mean

$$\sum_i w_i \mathbf{x}_i = \bar{\mathbf{x}} + \frac{1}{\varepsilon} (\Sigma \mathbf{y} - \mathbf{c}) + O(\varepsilon^{-2}), \quad \mathbf{c} = \frac{1}{2n} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}}) (\|\mathbf{x}_i\|^2 - \overline{\|\mathbf{x}\|^2})$$

$\Sigma = \frac{1}{n} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, the covariance of the training windows

1 $w_i \propto \exp(b_i/\varepsilon), \quad b_i = \mathbf{y}^\top \mathbf{x}_i - \frac{1}{2} \|\mathbf{x}_i\|^2 \quad (\|\mathbf{y}\|^2 \text{ cancels})$

2 $w_i = \frac{1}{n} (1 + (b_i - \bar{b})/\varepsilon) + O(\varepsilon^{-2})$

3 $\sum_i w_i \mathbf{x}_i = \bar{\mathbf{x}} + \frac{1}{n\varepsilon} \sum_i (b_i - \bar{b})(\mathbf{x}_i - \bar{\mathbf{x}}), \quad \text{substitute } b_i - \bar{b}$

checked: error $9.1 \times 10^{-4} \rightarrow 9.3 \times 10^{-6}$ as $\varepsilon: 10^3 \rightarrow 10^4$

Summary: the belief and the noise level are separate choices

noise ε	one Gaussian $\mathcal{N}(0, C)$	zero-variance mixture on training windows
small	ranking by $\ C^{-1}\mathbf{y}\ $	local scorer (1-NN distance)
$\approx \lambda_{d+1}$	global scorer (exact)	softmax-weighted local average
large	$\ \mathbf{y}\ $	$\ \mathbf{y} - \bar{\mathbf{x}}\ $

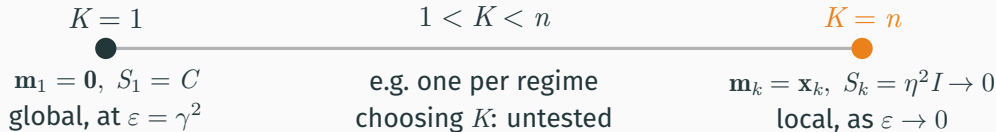
changing ε moves within a column, never across

checked by the scale sweep of the second column

One family contains both

$$\mu = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{m}_k, S_k) \Rightarrow \mathbb{E}[X | \mathbf{y}] = \sum_{k=1}^K r_k(\mathbf{y}) \left(\mathbf{m}_k + S_k (S_k + \varepsilon I)^{-1} (\mathbf{y} - \mathbf{m}_k) \right)$$

$$r_k(\mathbf{y}) \propto \pi_k \mathcal{N}(\mathbf{y}; \mathbf{m}_k, S_k + \varepsilon I)$$



recalled, not re-checked

mixtures of probabilistic PCA (Tipping & Bishop)

What is new and what is not

result	status	source
Tweedie's formula	classical	Efron 2011
Gaussian belief = soft projector	likely known	Wiener filter, probabilistic PCA
mixture belief on training windows	special case	manifold Parzen windows
one account of both detectors	ours	both decks

ours: the assembly, the exact conditions, the checks